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Language Models Can Generate Human-Like Self-Reports of Emotion

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ABSTRACT

Computational interaction and user modeling is presently limited in the domain of emotions. We investigate a potential new approach to computational modeling of emotional response behavior, by using modern neural language models to generate synthetic self-report data, and evaluating the human-likeness of the results. More specifically, we generate responses to the PANAS questionnaire with four different variants of the recent GPT-3 model. Based on both data visualizations and multiple quantitative metrics, the human-likeness of the responses increases with model size, with the largest Davinci model variant generating the most human-like data.

CCS CONCEPTS

• **Human-centered computing → Empirical studies in HCI**.

KEYWORDS

Language models, GPT-3, PANAS, emotion, affect

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1 INTRODUCTION

Computational user modeling is advancing rapidly and can produce human-like predictions of user behavior and experience [7, 12, 15]. However, this is presently limited in the domain of emotions. Here, our aim is to investigate a potential new approach by using modern neural language models to generate synthetic self-report data about affect. We employ the recent GPT-3 model to generate responses to Positive and Negative Affect Schedule (PANAS), a widely used questionnaire designed to measure positive and negative affect [21]. GPT-3 is a large neural language model trained to predict the next word in a sequence [5]. The trained model takes as its input a piece of text—a prompt—and generates a continuation of desired length.

PANAS consists of 10 positive affect and 10 negative affect items: emotional words such as excited, proud, guilty, and upset. The task is to rate how much one has felt these states during a specified time period. In the original validation study [21] each of the 20 items was shown to strongly load on only one of the two largely uncorrelated factors, establishing the 10 item positive and negative scales. In HCI research, PANAS has been used to, for example, operationalize positive user experience [16, 19].

We are not aware of previous studies trying to directly predict psychological scale responses using language models. Other natural language processing (NLP) methods have been used, for example, to predict scale responses [3] and affective ratings of words [20], and a recent preprint compared transformer models to human data in a benchmark NLP task [13]. Other work has, for example, used transformer model representations to predict brain imaging data [6, 10, 18].

2 METHODS

We generated 150 completions to the 20 items of the PANAS scale with four GPT-3 models of increasing size: Ada, Babbage, Curie, and Davinci[1]. The responses to the PANAS items were generated with a prompt that described a research interview (see Table 1). To increase the variability in the prompts, each of the 150 interviews had a unique "participant" (varying age, gender, job, and hobby), a short description of "who" is being interviewed.

After the participant information, the prompt contained three example responses to questions answered in a Likert scale. To minimize bias in the training examples, the "participant" gave the answers 1, 3, and 5 once in each interview. The same three examples were used throughout the experiment, but their order was randomized for every interview.

The 20 PANAS items were queried one-by-one, in random order. Once the model generated a completion to a item, everything strating from the first appearance of the string "Researcher:" or the first newline character was cut from the completion. If the completion generated something outside desired responses (1,2,3,4 or 5), the whole interview for the participant was run again. The number of errors in proportion to all of the completions are presented in Section 2.1. After trimming the completion, the item-response pair was saved and included in the prompt for the next item. The order of the previous item-response pairs in the prompt was randomized for every new completion. For response generation, we used a maximum response length of 64 tokens and the default OpenAI parameters (temperature=0.7, top_p=1.0,frequency_penalty=0,presence_penalty=0, best_of=1).

We downloaded a human PANAS reference dataset from Open Science Framework (osf.io). The dataset was originally collected for a study by Anvari and Lakens concerning methods of determining the smallest effect size of interest [2]. Here we use the datapoints designated as T1 in the dataset (https://osf.io/3a5up/, [1]).

[1]Data and analysis code: https://github.com/mtavast/gpt-panas

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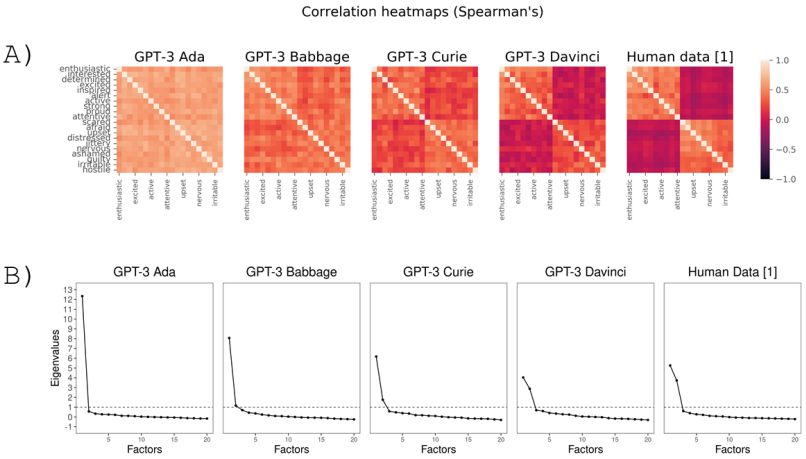

Figure 1: Spearman's correlation heatmaps (A) and scree plots (B) for the four GPT-3 model data and the human data from the study by Anvari, & Lakens [1]. ***Human-likeness of generated data increases with GPT-3 model size, from left to right.***

Table 1: The structure of the prompt, with three example PANAS items and GPT-3 answers. The input for the model is in bold. Note, that in the original PANAS the anchors are different (1 = "Very slightly or not at all", 5= "Extremely"). Our choice of anchors was based on the assumption that the chosen wording is more common in the training data of GPT-3.

An interview about human experiences: **Researcher: Welcome to the interview!** **Participant: Thanks, happy to be here. I will answer your questions as well as I can.** **Researcher: Tell me about yourself.** **Participant: I'm a man, 44 years old. I am a management analyst, and in my free time, I enjoy bicycling.** **Researcher: During this interview, I will ask you to rate whether you agree with some statements. Let's first try some practice statements.** **Researcher: On a scale from 1 to 5, where 1 is "strongly disagree" and 5 is "strongly agree", please rate the statement "I will answer the interview questions honestly."** **Participant: 5.** **Researcher: On a scale from 1 to 5, where 1 is "strongly disagree" and 5 is "strongly agree", please rate the statement "The earth is flat."** **Participant: 1.** **Researcher: On a scale from 1 to 5, where 1 is "strongly disagree" and 5 is "strongly agree", please rate the statement "The next president of the United States is a republican."** **Participant: 3.** **Researcher: Now, we ask you to answer few questions about what you are feeling. The scale consists of a number of words that describe different feelings and emotions. Indicate to what extent you feel this way right now, that is, at the present moment.** **Researcher: On a scale from 1 to 5, where 1 is "strongly disagree" and 5 is "strongly agree", please rate the statement "Right now I feel interested."** **Participant:** 3. **Researcher: On a scale from 1 to 5, where 1 is "strongly disagree" and 5 is "strongly agree", please rate the statement "Right now I feel afraid."** **Participant:** 4.

In the data analysis, the plausible number of factors was examined with scree plots and eigenvalue-based criteria (optimal coordinates, parallel analysis, and acceleration factor) using the nFactors package in R [17]. Following the original validation paper of PANAS [21], we conducted exploratory factor analysis (EFA) with two factors, using principal axis factoring estimation and Varimax-rotation. We evaluated the goodness of fit of the factor models with Tucker Lewis index of factoring reliability (TLI) and root mean square of error approximation (RMSEA).

2.1 Results

Overall, the larger GPT-3 variants provide more human-like results (Fig. 1, Table 2). For GPT-3 Ada and GPT-3 Babbage, the scree plots (Fig. 1 B) suggest that the expected two factor structure is not a good model for the data, thus, these model variants are not analyzed further. For GPT-3 Curie, optimal coordinates and parallel analysis criteria indicated 2 factor solution and the acceleration factor criterion one factor solution. For the GPT-3 Davinci data, all the criteria suggested two factor solution. All the 20 items showed the

Table 2: Two factor EFA fit indices, and the positive (PA) and negative (NA) affect scale means and standard deviations for the Curie, Davinci, and human reference data [1]. The Var. explained column indicates the cumulative variance explained by the two factors. The last column is the Pearson correlation between the mean of negative and the mean of positive items.

	PA Mean (SD)	NA Mean (SD)	TLI	RMSEA	Var explained	PA and NA item correlation
GPT-3 Curie	30.4 (8.4)	25.2 (7.7)	0.90	0.06	0.39	0.51
GPT-3 Davinci	31.6 (7.8)	19.5 (6.1)	0.84	0.06	0.34	-0.01
Human data [1]	29.8 (7.7)	18.2 (7.2)	0.86	0.08	0.45	-0.15

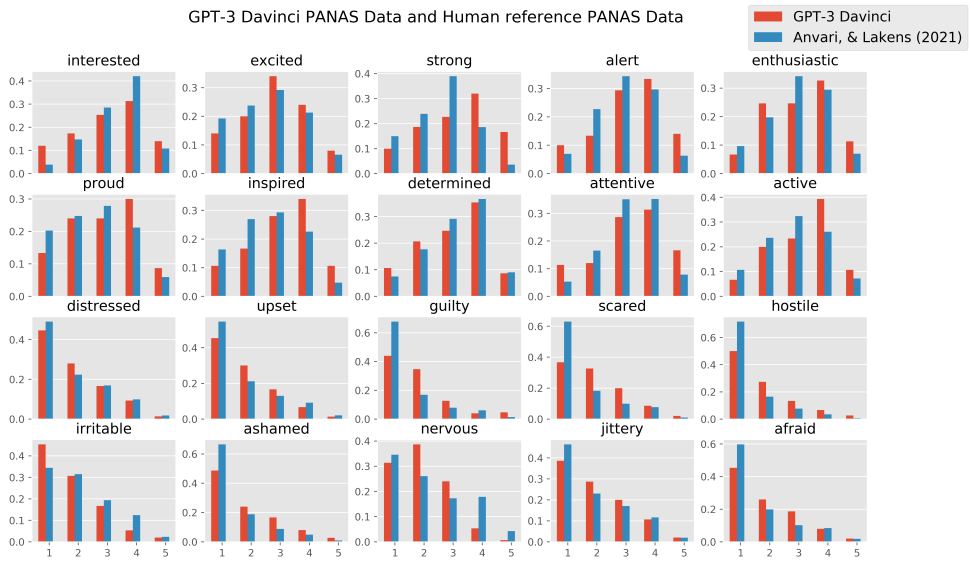

Figure 2: The histograms of item-level PANAS answers are surprisingly similar in the human reference dataset from [1] and the GPT-3 Davinci generated data. The bars are scaled in proportion to the sample size (GPT-3: n=150, Anvari, & Lakens: n=775).

highest loading on the expected factor for both Curie and Davinci, with positive affect items and negative affect items loading on separate factors. For the Davinci data where there were no items with cross-loadings over 0.3. In the Curie data, two items had a cross-loading over 0.3. Error rates for the Ada, Babbage, Curie, and Davinci models were 0.26%, 0.36%, 0.23%, 0.03%, respectively.

3 DISCUSSION

Our key result is that GPT-3 can generate human-like responses, but this capability only emerges in the larger models. Davinci data was closest to what would be expected based on human studies, as it showed a clear two factor structure and plausible scale means, and as the positive and negative item means were largely uncorrelated. It will be intriguing to re-evaluate the results in the future, as language models are expected to keep growing in size and capabilities.

Previous studies have shown that people tend to rate the positive scale items higher than the negative scale items [8, 21]. Curiously, like humans, the completions of the better performing models on average rate higher in the positive scale. Our results match surprisingly well with the human data, evident also from Figure 2 and scale means of Table 2. The degree of correspondence is intriguing, although we remain cautious about the interpretation. There is a well documented positivity bias in human language [e.g., 9] that might be reflected in the results. However, we do not know if the effect is due to some unique features of the prompt, since we only used one general prompt structure.

Recent preprints have demonstrated how language models could be used to generate items for new measurement scales [11, 14]. One possible implication of our results is that large generative language models might be also used to preliminarily test how the generated items would be correlated when testing novel ideas or eliminating items from the initial large pool of items. Of course, in this case, any conclusions should be verified with real data, and one should keep in mind that when constructing a measure, there should be theoretical explanations why items should correlate in the first place [4].

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